

AS/400 → Snowflake Cutover Runbook

Purpose: Ensure a controlled, auditable, zero-defect transition from AS/400 (DB2/400) to Snowflake.

1. Pre-Cutover Preparation (T-7 to T-1 Days)

1.1 Final Readiness Checks

- Environment readiness validated
- Landing zone storage verified
- Snowpipe & COPY pipelines tested
- Streams & Tasks running without lag
- DBT jobs passing all tests
- All reconciliation reports green

1.2 Business & Operations Readiness

- Business SMEs confirm validation windows
- Downstream systems prepared for connection string updates
- Support teams staffed for 24×7 coverage
- Communication plan distributed

1.3 Cutover Command Center Setup

- Slack/Teams war room created
- On-call rotation assigned
- Escalation matrix activated

2. Freeze Window (T-0 Day, Start of Cutover)

2.1 AS/400 Freeze Activation

- Job scheduler freeze initiated

- No structural changes allowed on PF/LF files
- No new RPG/CL deployments
- Business notified of freeze start

2.2 Confirm No In-Flight Jobs

- Validate no long-running AS/400 jobs
- Validate no open transactions
- Validate no pending journal rollovers

3. Final Full Load + CDC Replay (T-0)

3.1 Final Full Extract

- Execute final PF/LF unload scripts
- Validate row counts, checksums, timestamps
- Land files in raw zone

3.2 Final COPY INTO Loads

- Load final full extracts into Snowflake
- Validate ingestion row counts
- Validate rejected rows = 0

3.3 Final CDC Replay

- Journaling-based CDC replayed into Snowflake
- Streams & Tasks merge final deltas
- Validate CDC lag = 0

4. Cutover Validation (T-0)

4.1 Schema Validation

- Schema checks

- Column counts
- Data types
- Nullability

4.2 Metric Validation

- Row counts, sums, distincts
- Compare AS/400 vs Snowflake

4.3 Row-Level Validation

- Hash-based reconciliation
- Exceptions reviewed and approved

4.4 Functional Validation

- Legacy vs Snowflake output comparison
- Validate business logic rebuilt in DBT
- Validate SCD Type 1/2 behavior

4.5 Business Sign-Off

- SMEs validate critical reports
- Business approves cutover continuation

5. Switch-Over (T-0)

5.1 Downstream System Cutover

- Update BI tools (Power BI, Tableau, MicroStrategy)
- Update APIs and integration endpoints
- Update ETL/ELT pipelines to Snowflake

5.2 Activate Snowflake as System of Record

- Disable AS/400 data feeds
- Enable Snowflake feeds
- Validate downstream consumption

5.3 Enable Monitoring

- Query history monitoring
- Warehouse utilization
- Snowpipe ingestion health
- DBT job monitoring

6. Post-Cutover Hypercare (T+1 to T+14 Days)

6.1 Daily Validation

- Row counts
- CDC deltas
- Snowpipe ingestion
- DBT job success rates

6.2 Performance Monitoring

- Warehouse right-sizing
- Clustering key effectiveness
- Query performance tuning

6.3 Issue Management

- Severity 1 issues escalated immediately
- Severity 2–4 logged and triaged
- Daily stand-ups with business

6.4 Business Acceptance

- Final reconciliation
- Final sign-off
- Transition to steady-state operations

7. Decommissioning (Post Hypercare)

7.1 AS/400 Decommissioning

- Retire PF/LF files
- Disable RPG/CL jobs
- Shut down LPARs
- Archive historical data

7.2 Snowflake Optimization

- Cost optimization
- RBAC/ABAC hardening
- Storage tiering
- Warehouse tuning

8. Cutover Checklist (Executive Summary)

Pre-Cutover

- Environments ready
- Pipelines validated
- SMEs aligned

Freeze

- AS/400 locked
- No in-flight jobs

Final Loads

- Full load
- CDC replay
- Ingestion validated

Validation

- Schema
- Metrics
- Row-level
- Functional

Switch-Over

- Downstream systems updated
- Snowflake activated

Hypercare

- Daily monitoring
- Issue resolution
- Business acceptance